Reg. No: 192525370

SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Name of the Student : Dharshan .A.N

Programme: **B.E / B.Tech**

Slot : A

Course Code: **CSA15**

Faculty : Dr.Arul Raj A.M

Course Name: **Cloud Computing and Big Data Analytics**

Date: 02-11-2026

Duration: **1:30 Hr**

Max Marks: **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions.
CO2	Analyze virtualization architectures to identify performance and security issues in cloud computing environments.

Overall Marks Distribution :

Question No	Question	CO	Bloom's Level	Marks
1	Q1 – Cloud Evolution and Service Models	CO1	BL3	10
2	Q2 – Server Consolidation Energy Saving	CO1	BL3	10
3	Q3 – VM Over-Commitme Security	CO2	BL4	10
4	Q4 – Data Centre, SDDC, and Privacy	CO2	BL4	10
5	Q5 – Edge Filtering and Serverless Computing	CO2	CO2	10
	Total			50

Student Signature	Faculty Signature	Program Director

CIA 1 TEST

Answer all the Questions

1. A media company, StreamWave, plans to modernize its IT environment. It wants (i) to move an existing content-management application to virtual machines while retaining control over the operating system and installed software, (ii) to develop and deploy a new customer-facing application without maintaining servers or runtime environments, and (iii) to subscribe to a ready-made customer relationship management (CRM) application through a browser. Explain how developments in computing hardware and Internet software have enabled this transition. Map the three requirements to IaaS, PaaS and SaaS, respectively, and explain how the broader XaaS model can support additional requirements. Justify your choices.

(10) (CO1) (BL3)

2. A logistics company operates 15 physical servers. Each server has an average CPU utilisation of 25% and consumes 350 W. The workloads will be consolidated onto new virtualization hosts, each consuming 450 W. The target average CPU utilisation is 65% per host. The company requires N+1 fault tolerance (one additional host beyond the calculated capacity). Electricity costs ₹9 per kWh, and the systems operate 24 × 365 hours per year. Calculate (a) the number of active hosts required and the total hosts including N+1, (b) the consolidation ratio, (c) utilisation when all hosts are active and after one host fails, and (d) the annual electricity cost saving. State your assumptions.

(10) (CO1) (BL3)

Given data

Parameter	Value
Physical servers	15
Average utilisation per server	25%
Power per old server	350 W
Power per new host	450 W
Target utilisation per host	65%
Fault tolerance	N+1 (one extra host)
Electricity tariff	₹9 per kWh
Operating hours per year	24 × 365 = 8,760 h

(10) (CO1) (BL3)

3. A virtualization host has 24 physical CPU cores and 96 GB RAM, of which 8 GB is reserved for the hypervisor. Twelve VMs are deployed, each configured with 3 vCPUs and 8 GB RAM. During normal operation, each VM uses an average of 5 GB RAM and 35% of its allocated vCPU capacity. During peak operation, eight VMs use 8 GB RAM each and the remaining four use 6 GB each; all VMs use 75% of their allocated vCPU capacity. Analyse (a) the vCPU and memory over-commitment ratios, (b) average and peak memory sufficiency, (c) average and peak CPU sufficiency, and (d) possible performance and security issues with suitable remedies.

(10) (CO2) (BL4)

4. Two independent universities operate separate cloud data centres and propose to establish a cloud federation to share research-computing and digital-library services. Analyse how a conventional cloud data-centre architecture and a Software-Defined Data Centre (SDDC) can support this arrangement. Explain how Service-Oriented Architecture (SOA) enables service sharing between the universities. Examine the federation challenges and the privacy risks

associated with user presence information and federated identity. Recommend suitable technical and administrative safeguards.

(10) (CO2) (BL4)

5. A public-transport authority operates 150 road-monitoring cameras. Each camera generates one 4 MB video clip every 3 minutes, continuously. All clips are uploaded to cloud object storage through a REST API, and one serverless function is invoked for every uploaded clip. Assume 1 GB = 1000 MB, a 30-day month, a storage price of ₹1.50 per GB-month and a serverless charge of ₹0.00015 per invocation. An edge-filtering system is proposed to upload only 25% of the clips. Calculate (a) the daily data volume and average bandwidth without filtering, (b) the 30-day storage cost without filtering, (c) the daily data volume, average bandwidth, storage cost and monthly saving after edge filtering, and (d) the monthly serverless invocation cost after filtering. Explain the role of the REST API and why object storage is suitable for video clips.

(10) (CO2) (BL3)

Given data

Parameter	Value
Cameras	150
Clip size / interval	4MB every 3 minutes
Storage price	₹1 per GB-month
Edge filtering	25% of clips uploaded
Serverless price	₹0.0002 per invocation
Month / unit	30 days; 1 GB = 1000 MB

① Media Company

Cloud management is way of modernizing the share of resource through internet. it provides computing resources like servers, storage and software.

Hardware & Software Development:

* Visualization enables multiple virtual machine in one server.
* powerful processor improve performance.
* High-speed internet enables cloud access.

Iaas [Infrastuse as a Service]:

* Users for ingraing existing contruct-management application.
* Uses law Control os & installed software.
* Eg: Amazon EC2

Paas [Platfoom as a Service]:

* Used for developing & deploying new applicatio
* Used to usable servers.
* Eg: Google App Engine

SaaS [Software as a Service]:

* Used for accessing ready made cpu software.
* provides handles updates & maintence.
* Eg:- Salesforce.

XaaS [Everything as a Service]:

* STaaS - Storage as a Service.
* DBaaS - Database as a Service
* NaaS - Network as a Service
* SECaaS - Security as a Service
* BaaS - Backup as a Service.

Conclusion:

IaaS, PaaS & SaaS provide information develop-ent platforms & ready made software XaaS offers additional cloud services. reducing cost & improving scalability.

② Logistics Company :

Given:

Physical server = 15

CPU utilization = 25%

Power per server = 350W

New power host = 450W

Target Utilization = 65%

Sol^n :-

a) Number of hosts:

Total workload = $15 \times 25\%$ = 3.75 $\quad \frac{15 \times 25}{100}$

Active hosts = 3.75 / 0.65 = 5.77 = ~ 6 hosts

Including =N+1 = 6 + 1 = 7 hosts

b) Consolidation ration

15/7 = 2.14 = 1

c) CPU Utilization :-

* All 7 hosts active = (3.75/7) X100 = 53.57%

* On host fails = 3.75/6 X100 = 62.5 %

d) Annual electricity savings

Old power = 15 X350 = 5250W

New power = 7X450 = 2150W

Power Saved = 2100 W = 2.1 kW

Annual Energy Saved = 2.1 × 8760

= 18396 kWh

Cost Savings: 18396 × ₹9

= ₹1,65,564 per year.

3) Virtualization host:

Given:

Physical CPU = 24

Total ram = 96 GB

Hypervisor RAM = 8 GB

VMs = 12

Each VM = 3vCPUs and 8GB RAM

a) Over-Commitment ratios:

Total vCPU = ~~96~~ 8 12 × 3 = 36.

CPU over Commitment = 36/24 = 1.5 : 1

Usable RAM = 96 - 8 = 88 GB

Allocated RAM = 12 × 8 = 96 GB

Memory over-commitment = 96/88 = 1.09 : 1

b) Memory Sufficiency :-

Average memory = 12 x 5 = 60 GiB

Peak memory = (6 x 8) + (1 x 6) = 88 GiB

*Average : Sufficient

*Peak : Sufficient but no free space remains.

c) CPU ~~sufficient~~ sufficiency

Average CPU demand = 36 x 35% = 12.6 cores

Peak CPU demand = 36 x 75% = 27 cores

*Average : Sufficient (12.6 Cr)

*Peak : Insufficient (27.7 24)

d) Issues and Remedies

Issues :-

* CPU ~~content~~ condition and performes reduces.

* Memory exhaustions

Remedies :-

*Add more physical CPU resources.

*monitor and optimize VM allocation.

*Use network ~~allocation~~ isolation & encryption.

~~4) IaaS, PaaS, SaaS & XaaS.~~

4) Two Independent Universities

Cloud federation

* It provides servers, storage and networking.
* It also improves resource utilization.

SDDC (Software-Defined Data Centre):

* It uses software to manage infrastructure.
* Automates resource allocation.

Role of SOA (Service Oriented Architecture):

* Enables service sharing between universities
* Supports visitor operability.

Challenges:-

* Facilities management between universities.
* User present information leakage.
* Data privacy & Security.
* Resource Sharing & Utilization.

Safeguard:

* Implement federated authentication and SSO.
* Use encryption & role-base access.
* Apply firewalls & acess policies
* Establishes data-sharing arguments.

Conclusion :-

Although, having a centralised cloud federation to govern the law and establish rules over the cloud managment, there is always a risk malware and maintence to the server which could shut down whole cloud federation.

5) Public Transport :

- Cameras = 150
- clip size = 4 MB
- one clip every 3 minute
- Storage cost = £1.50 per GB . months
- Serverless cost = £0.0002 per invocation

a) Without edge filtering :

clips per camera = 480

Total clips = 150 × 480 = 72,000

Daily data = 72,000 × 4 = 288,000

UB = 288 GB

Average bandwidth = 288 × 8 / 86,100 = 0.2667 Gbps

6) Monthly Storage cost :

30-day storage = 288 × 30 = 8640 GB

cost' = 8640 × 1.50 = £12,960

c) Width 25% edge Filtering:

Daily data = 288 × 25% = 72 GB

Average bandwidth = 6.67 Mbps.

monthly storage = 72 × 30 = 2160 GB

Storage cost = 2160 × 1.50 = 3240

d) Serveless Invocation cost:

Daily invocation = 72,000 × 25% = 18,00

monthly invocation = 18,000 × 30 = ₹ 40,000

Cost = ₹ 40,000 × 0.00015 = 281